

Gyrokinetic Simulations of Microtearing Mode in 2D Slab Model



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Introduction

- Normal tearing mode can be unstable when magnetic shear exists in magnetized plasma [Furth+, 1963].
- Microtearing mode (MTM) can be unstable even in the regime where normal tearing mode is stable ($\Delta' < 0$) [Gladd+, 1980].
- MTM is driven by electron temperature gradient [Hazeltine+, 1975, Drake+, 1977, Gladd+, 1980].
- MTM is believed to be cause of electron thermal transport in recent tokamak devices [Kotschenreuther+, 2000].
- The perturbation spectrum similar to MTM was observed in NSTX and MAST devices [Gutenfelder+, 2011].
- Some gyrokinetic simulation studies suggest the existence of collisionless MTM in a toroidal geometry due to trapped particles [Chowdhury+, 2015], and even in a slab [Kobayashi+, 2014].
- Theoretical study based on the reduced gyrokinetic model claims the energy dependent collision model is not necessary to destabilize MTM [Zocco+, 2015].
- In this study, we perform gyrokinetic simulation of MTM using AstroGK and show how the out-of-plane electric field is generated in terms of electron dynamics in the current layer.

Problem Setup

We perform linear simulation by using gyrokinetic simulation code AstroGK [Numata+, 2010].

- 2D slab ($k_z = 0$).
- Initial magnetic configuration : $B_0 = B_{z0}\hat{z} + B_{y0}^{\text{eq}}(x)\hat{y}$, ($B_{z0} \gg B_{y0}^{\text{eq}}(x)$)
- We give $B_{y0}^{\text{eq}} = B_{y0}\sin(x/a)$, hence Tearing stability parameter is given by

$$\Delta' a = \begin{cases} 2\sqrt{1 - (k_y a)^2} \tan \frac{\pi}{2} \sqrt{1 - (k_y a)^2} & , 0 < k_y a < 1 \\ -2\sqrt{(k_y a)^2 - 1} & , k_y a > 1 \end{cases}$$
 where a is magnetic shear length at $x = 0$.
- Electron temperature and density gradients :

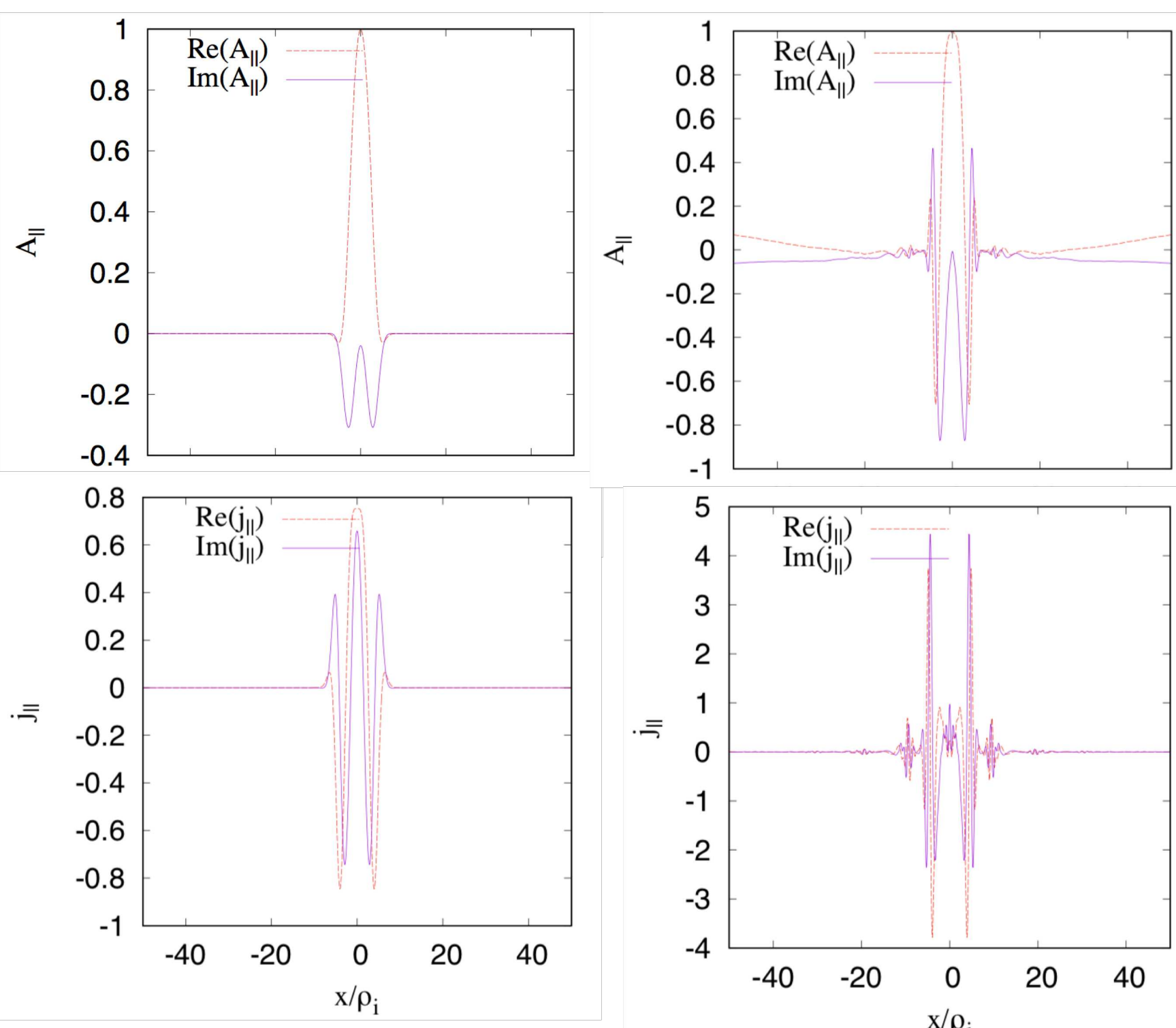
$$\begin{cases} L_{T0e}^{-1} = -\partial(\ln T_{0e})/\partial x \\ L_{n0e}^{-1} = -\partial(\ln n_{0e})/\partial x \end{cases}$$
- We define diamagnetic drift and the drift frequency.

$$U_{T0e,n0e}^d = T_{0e}/(eB_{z0}L_{T0e,n0e}) \quad \omega_{T0e,n0e}^* = k_y U_{T0e,n0e}^d$$

Eigenfunctions of Microtearing Mode

Case1: Unstable

Case2: Stable



The structures of MTM show a tearing parity (even parity). The parallel current (bottom) is localized within a narrow layer. As ν_{ei}/ω_e^* becomes weaker, $j_{\parallel}$ becomes more narrow. The results show qualitative agreement with the previous theoretical works [Gladd+, 1980, Zocco+, 2015].

Evaluation of parallel electric field in current layer

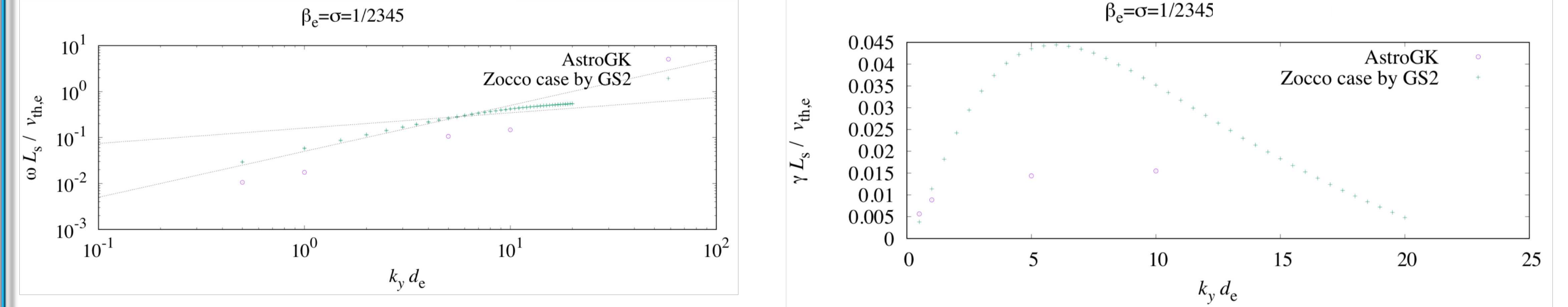
- From induction equation, the magnetic field diffusion is occurred by parallel electric field in current layer. Since MTM is the mode with magnetic reconnection, it is necessary to investigate the generation mechanism of the electric field.
- Focus on Ohm's law: $\tilde{E}_{\parallel} = -i\omega_{T0e}^* \tilde{A}_{\parallel} - \frac{T_{0e}}{q_e} i k_y f(x) \left(\frac{\delta \tilde{n}_e}{n_{0e}} + \frac{\delta \tilde{T}_{\parallel e}}{T_{0e}} \right) - \eta j_{\parallel} + i\omega_{d_e}^2 \tilde{j}_{\parallel} = E_4$

(Where $\tilde{E}_{\parallel} = i\omega \tilde{A}_{\parallel} - i k_y f(x) \tilde{\phi}$, $\eta = \nu_{ei} d_e^2$ and $f(x)$ means antiparallel magnetic field.)
- The eigenfunction $\tilde{E}_{\parallel} = i\omega \tilde{A}_{\parallel} - i k_y f(x) \tilde{\phi}$ (right) is shown. Since $\tilde{E}_{\parallel} = -\gamma \tilde{A}_{\parallel}$ at $x = 0$, $\tilde{E}_{\parallel}(x=0) < 0$ in case 1 and $\tilde{E}_{\parallel}(x=0) > 0$ in case 2.
- Case 1: E1 (diamagnetic drift term) contributes to stabilization, E3 (electric resistivity) and E4 (electron inertia) contribute to destabilization.
- Case 2: Not only E1 but also E3 contributes to stabilization as collision weakens.
- E2 is sufficiently smaller than the other terms in both Case1 and Case2.

Simulation Results

Comparison with Linear Theory [Zocco+, 2015]

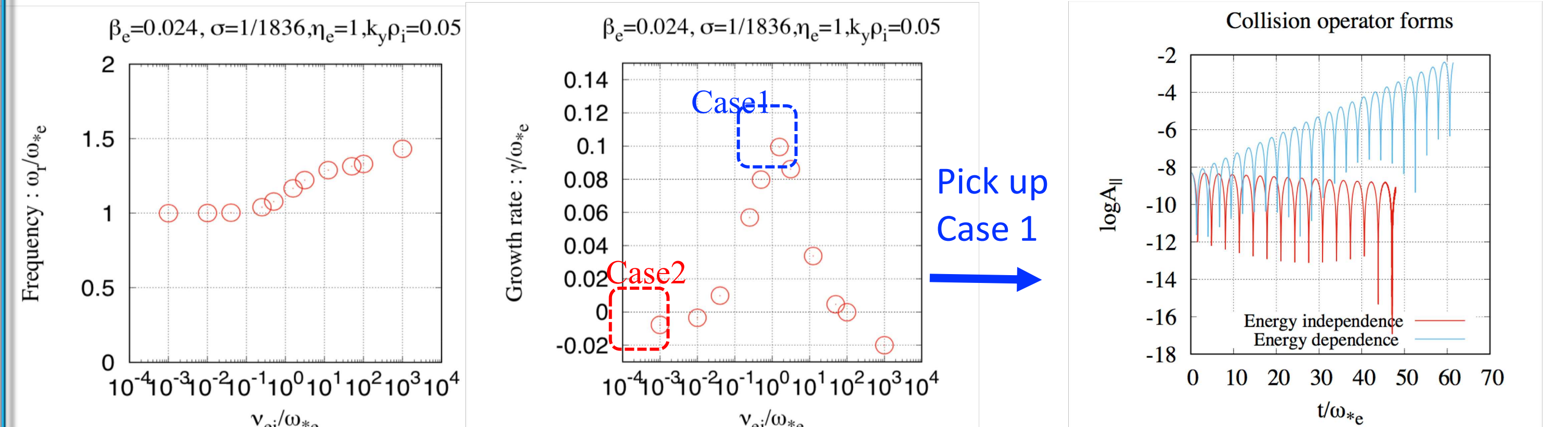
- We have firstly compared our simulation results with linear theory of [Zocco+, 2015] in the low-beta and weakly collisional regimes.



- Give magnetic shear and electron temperature and density gradients in low-beta and collisionless regimes ($L_{T0e}^{-1} \gg L_{n0e}^{-1}$).
- GS2 and AstroGK differ in the way magnetic shear is given (given by sinusoidal in AstroGK, while given by linear function in GS2).
- We give the ratio of shear length and electron temperature gradient : $aL_{T0e}^{-1} \approx 270$ in AstroGK, while the ratio is given by $aL_{T0e}^{-1} \approx 27$ in GS2. The other parameters are matched in AstroGK and GS2.
- A qualitative agreement between linear theory and our result have been obtained.

Collision frequency dependence

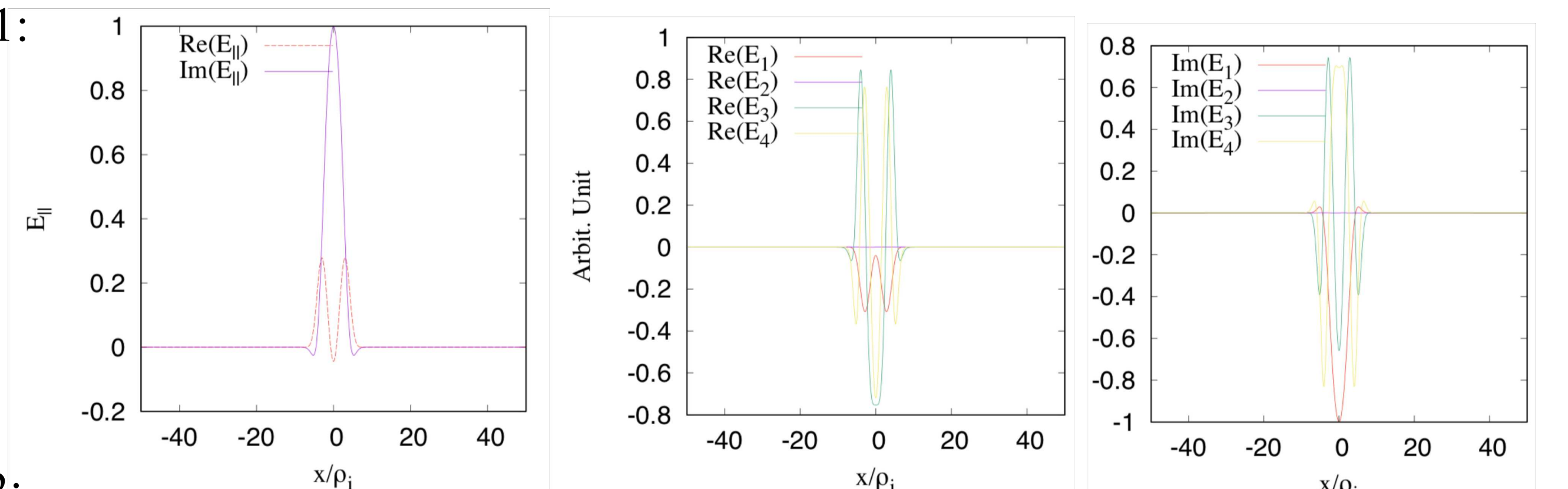
Parameters : $\beta_e = n_{0e} T_{0e} / (B_{z0}^2 / 2\mu_0) = 0.024$, $\sigma = m_e / m_i = 1/1836$, $n_{0e} / n_{0i} = 1$, $\tau = T_{0i} / T_{0e} = 1$, $\Delta' a = -4.58$



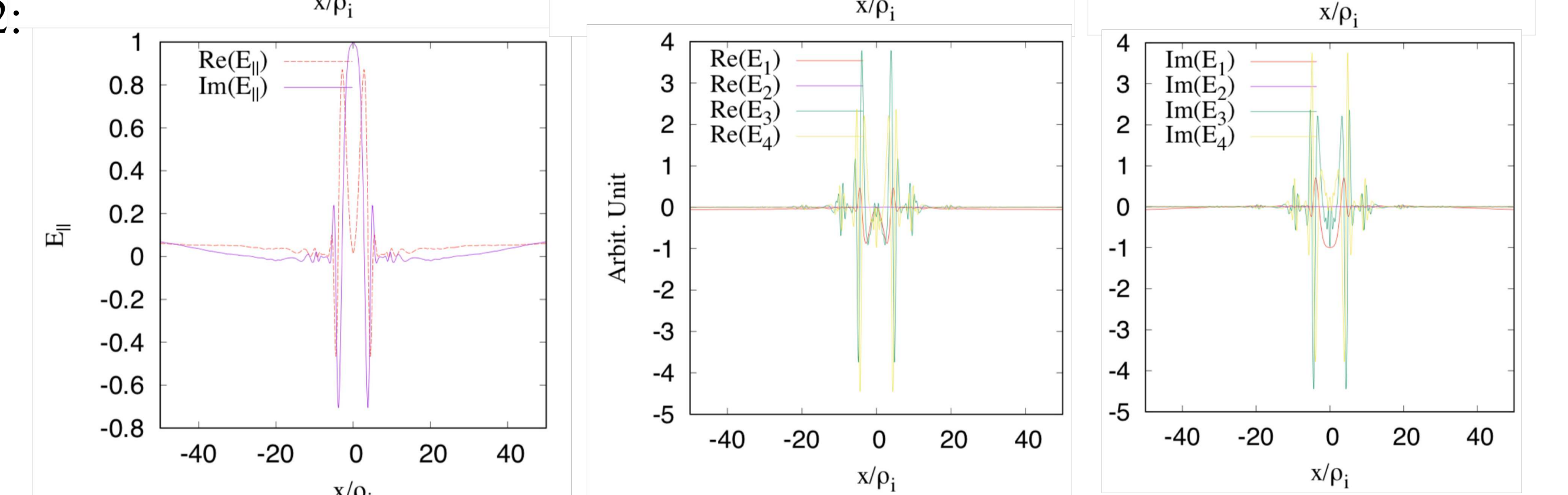
- Growth rate (right) strongly depends on electron-ion collision. γ/ω_e^* peaks at $\nu_{ei} \sim \omega_e^*$ and the mode is stable in both weakly and strongly collisional regimes.
- Pick up Case 1 and when the energy dependent collision model (A) and the energy independent one (B) are compared, MTM is stable in (B).

- Since E3 due to electron-ion collision is necessary to destabilize MTM, the mode is the collisional mode.

Case1:



Case2:



Conclusions

- We have performed the linear simulations of microtearing instability by using AstroGK and investigated collision dependence of MTM.
- The energy dependent collision model is necessary to destabilize MTM in AstroGK.
- The structures of MTM is localized within narrow layer and show qualitative agreement with the previous studies.
- We have investigated the parallel electric field in current layer in terms of Ohm's law.
 - Electric resistivity due to electron-ion collision is necessary to destabilize MTM.

Microtearing Mode is collisional mode !

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